



Lab 1 Report

System Identification

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1 Introduction

The aim of this laboratory was to identify a workable dynamic model of the suspension simulator using two complementary experiments: a step-response test and a frequency-response test. The laboratory system is a simulated spring–mass–damper system whose parameters are unique to the student. The lab brief specifically requires the time-domain response to be used to estimate the gain, pole type and dominant pole location, and then requires sinusoidal tests to confirm those estimates in the frequency domain [1].

The measured system was found to be a lightly overshooting but relatively well-damped second-order system. The step test was first used to estimate the steady-state gain, damping ratio, natural frequency and complex pole pair. Nine sinusoidal tests were then performed between 0.500 rad s^{-1} and 8.000 rad s^{-1} , with extra points placed around the frequency predicted by the step test. Finally, the independently estimated time- and frequency-domain models were combined and validated against both experimental datasets. The report includes a linked table of contents, list of figures and list of tables to make the experimental evidence and calculations easy to navigate.

2 System Modelling

2.1 Physical model

The simulated suspension is described in the lab brief by

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t), \quad (1)$$

where $x(t)$ is displacement, b is the damping coefficient, k is the spring coefficient, and $F(t)$ is the applied force. The lab specifies that the parameters are scaled to the mass, so $m = 1$, and that the applied force is directly proportional to the simulator voltage input [1]. Let the proportionality constant between simulator input $u(t)$ and force be c , so that $F(t) = cu(t)$. Equation (1) becomes

$$\ddot{x}(t) + b\dot{x}(t) + kx(t) = cu(t). \quad (2)$$

Taking the Laplace transform with zero initial conditions gives

$$G(s) = \frac{X(s)}{U(s)} = \frac{c}{s^2 + bs + k}. \quad (3)$$

For identification, it is convenient to write the same transfer function in the standard second-order form

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad (4)$$

where K is the DC gain, ζ is the damping ratio, and ω_n is the natural frequency. Comparing Equations (3) and (4) gives

$$b = 2\zeta\omega_n, \quad k = \omega_n^2, \quad c = K\omega_n^2. \quad (5)$$

2.2 Experimental method

The suspension simulator was operated in “Laboratory Testing” mode. The manual states that the logged input and output displacement are sampled every 100.000 ms [2]. The recorded CSV files were analysed as follows:

1. For the step test, the step instant was shifted to $t = 0$ and the initial displacement was subtracted from the output. The actual recorded input step was $\Delta u = 4$, so all gain calculations were normalised by this known step size rather than assuming a unit step.
2. For the frequency test, sinusoidal tests were performed at nine angular frequencies from 0.5 to 8 rad/s, with additional measurements around the predicted corner region. The measured input was approximately a sinusoid of amplitude 2 about a DC offset of 4. The offset does not affect the sinusoidal gain calculation for an LTI model; only the AC amplitudes were used.
3. At each test frequency, the first four cycles were discarded to reduce transient contamination. A sine/cosine least-squares fit was then used to determine the input and output amplitudes and phase angles.
4. The resulting model was validated against both the measured step response and the measured frequency response, as required by the marking guide.

The two experiments and the information extracted from each are summarised in [Table 1](#).

Table 1: Experimental test plan and identification purpose.

Test	Applied input	Main quantities obtained
Step response	$0 \rightarrow 4$ constant input	DC gain, overshoot, damping ratio, peak time, natural frequency, pole pair, initial model
Frequency response	Nine sinusoids from 0.5 to 8 rad/s	Low-frequency gain, phase trend, natural-frequency region, high-frequency roll-off, independent model
Validation	Same measured datasets	Agreement of proposed model with both time- and frequency-domain measurements

3 System Identification

3.1 Step Response Test

[Figure 1](#) shows the recorded simulator input. In the raw log, the input changes from 0 to 4 at approximately $t = 16.300$ s. For analysis, this instant was shifted to $t = 0$. Because the step size is known, the use of a 4-unit step does not change the identification method; the steady-state output change is simply divided by four.

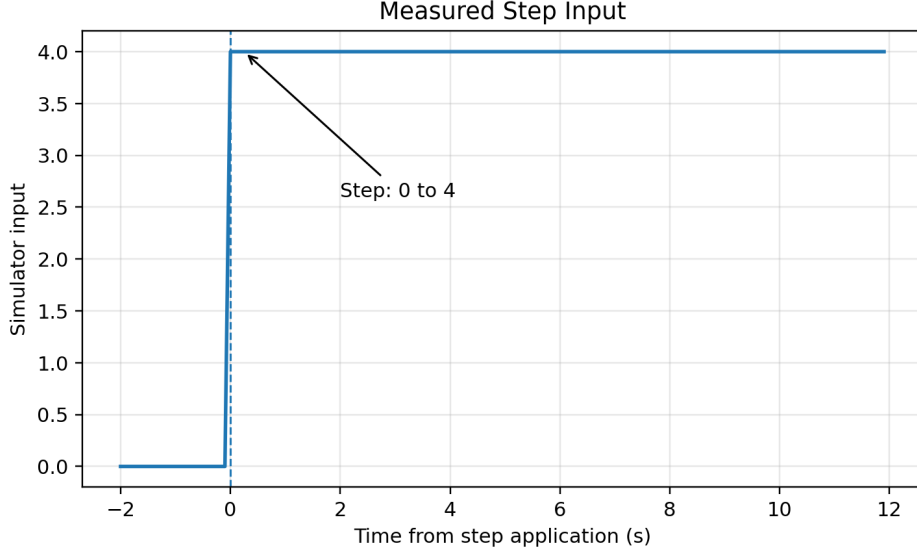


Figure 1: Measured step input. The recorded input changes from 0 to 4 at $t = 0$.

The initial displacement was 5.000 and the final displacement was 7.552, giving an output change

$$\Delta y_{ss} = 7.552 - 5.000 = 2.552. \quad (6)$$

The DC gain is therefore

$$K_{\text{step}} = \frac{\Delta y_{ss}}{\Delta u} = \frac{2.552}{4} = 0.638. \quad (7)$$

The principal measured values used in the step-response calculations are collected in [Table 2](#). This makes the origin of each numerical value explicit.

Table 2: Measured quantities from the step-response test.

Quantity	Measured value
Raw step time	16.3 s
Input change, Δu	$0 \rightarrow 4$
Initial displacement, y_0	5.000
Final displacement, y_{ss}	7.552
Steady-state displacement change, Δy_{ss}	2.552
First peak displacement, y_{peak}	7.628
Peak time relative to the step, t_p	2.6 s

The zero-adjusted response is shown in [Figure 2](#). A small but clear overshoot is present, which confirms that the poles are a complex-conjugate pair and that $0 < \zeta < 1$. In the raw log the first peak occurs at approximately $t = 18.900$ s; relative to the step at 16.300 s this gives $t_p \approx 2.600$ s. The corresponding peak displacement is 7.628, or a zero-adjusted displacement change of 2.628. Hence

$$M_p = \frac{2.628 - 2.552}{2.552} = 0.0298, \quad (8)$$

which corresponds to a percentage overshoot of approximately 2.98%.

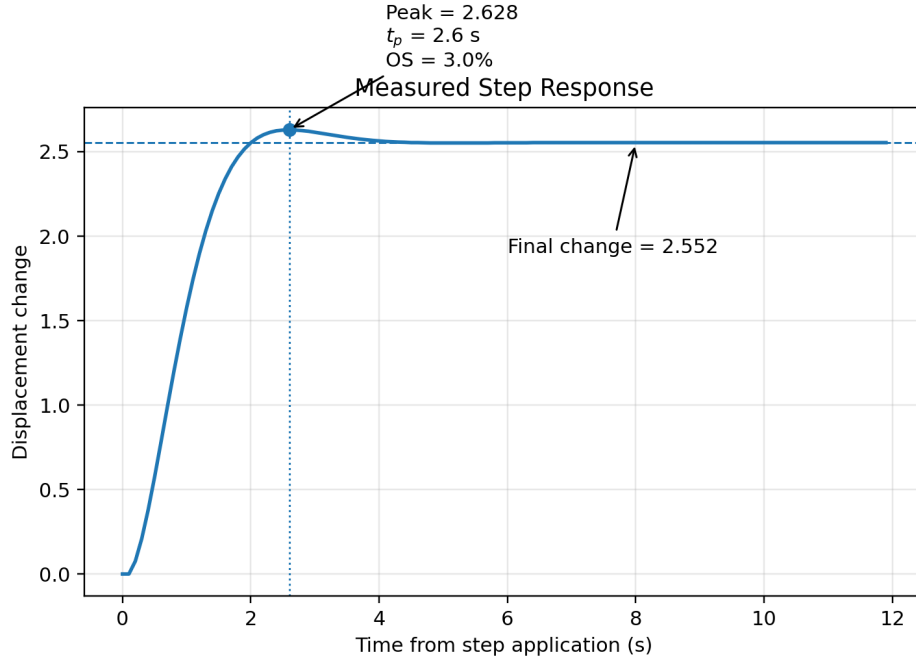


Figure 2: Measured zero-adjusted step response. The final displacement change, peak value, peak time and overshoot are annotated.

For an underdamped second-order system,

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}, \quad (9)$$

so the damping ratio can be calculated directly as

$$\zeta_{\text{step}} = \frac{-\ln(M_p)}{\sqrt{\pi^2 + [\ln(M_p)]^2}} = 0.745. \quad (10)$$

The damped natural frequency follows from the peak time:

$$\omega_d = \frac{\pi}{t_p} = \frac{\pi}{2.6} = 1.208 \text{ rad s}^{-1}. \quad (11)$$

Thus

$$\omega_{n,\text{step}} = \frac{\omega_d}{\sqrt{1-\zeta^2}} = 1.813 \text{ rad s}^{-1}. \quad (12)$$

The dominant complex pole pair is therefore

$$s = -\zeta\omega_n \pm j\omega_d = -1.351 \pm j1.208. \quad (13)$$

As a check, the measured 2% settling time is approximately 3.200 s. The usual estimate $4/T_s \approx 1.25 \text{ s}^{-1}$ is reasonably close to the calculated pole real part, $\zeta\omega_n = 1.35 \text{ s}^{-1}$.

Substitution into Equation (4) gives the step-test model

$$G_{\text{step}}(s) = \frac{2.097}{s^2 + 2.703s + 3.287}. \quad (14)$$

The corresponding parameter estimates are

$$b_{\text{step}} = 2.703, \quad k_{\text{step}} = 3.287. \quad (15)$$

Table 3 summarises the complete parameter set obtained from Stage 1.

Table 3: Parameters identified from the step response.

Parameter	Symbol	Estimate
DC gain	K_{step}	0.638
Percentage overshoot	$\%OS$	2.98%
Damping ratio	ζ_{step}	0.745
Damped natural frequency	ω_d	1.208 rad/s
Natural frequency	$\omega_{n,\text{step}}$	1.813 rad/s
Dominant poles	$s_{1,2}$	$-1.351 \pm j1.208$
Damping coefficient	b_{step}	2.703
Spring coefficient	k_{step}	3.287

The important frequency predicted from the step response is therefore around $\omega_n \approx 1.800 \text{ rad s}^{-1}$ (the relative -3 dB point of this model is approximately 1.710 rad s^{-1}). This estimate motivated the denser frequency sampling between 1.000 rad s^{-1} and 2.200 rad s^{-1} .

3.2 Frequency Response Test

The chosen frequencies span well below, around and well above the natural-frequency estimate from the step test. In particular, the points at 1.4, 1.8 and 2.200 rad s^{-1} provide extra resolution around the expected corner region, while 3, 5 and 8.000 rad s^{-1} reveal the high-frequency roll-off.

The choice of test points is summarised in Table 4; the denser set around 1–2.2 rad/s was deliberately selected because Stage 1 predicted an important frequency near 1.8 rad/s.

Table 4: Rationale for the selected sinusoidal test frequencies.

Frequency band	Test points (rad/s)	Purpose
Low frequency	0.5, 0.8	Estimate the low-frequency/DC gain before the main roll-off
Transition region	1.0, 1.4	Observe the start of the magnitude decrease and increasing phase lag
Predicted corner region	1.8, 2.2	Resolve the region around the step-test estimate $\omega_n \approx 1.8 \text{ rad/s}$
High frequency	3, 5, 8	Confirm the second-order roll-off and increasing phase lag

For each frequency, the sinusoidal input and output were represented as

$$q(t) = q_0 + a \sin(\omega t) + b \cos(\omega t), \quad (16)$$

from which the sinusoidal amplitude and phase are

$$A_q = \sqrt{a^2 + b^2}, \quad \phi_q = \tan^{-1}\left(\frac{b}{a}\right). \quad (17)$$

The frequency-response magnitude was then calculated from

$$|G(j\omega)| = \frac{A_{out}}{A_{in}}, \quad G_{dB} = 20 \log_{10} |G(j\omega)|, \quad (18)$$

and the phase lag was obtained from $\phi_{out} - \phi_{in}$.

The measured magnitude and aligned phase values are listed in [Table 5](#). The raw phase measurements contained an additional frequency-proportional lag very close to $\omega(0.1 \text{ s})$. Since the manual specifies a 100.000 ms logging interval [\[2\]](#), this was treated as a one-sample alignment effect rather than part of the continuous-time plant. The plotted phase values therefore use the correction

$$\phi_{aligned} = \phi_{raw} + \omega(0.1) \frac{180}{\pi}. \quad (19)$$

This correction makes the measured phase consistent with the simultaneously measured magnitude and the step-response dynamics.

Table 5: Measured frequency-response data.

ω (rad/s)	$ G(j\omega) $	Magnitude (dB)	Aligned phase (deg)
0.5	0.631	-4.00	-23.5
0.8	0.615	-4.22	-38.4
1.0	0.595	-4.51	-48.7
1.4	0.530	-5.52	-69.3
1.8	0.441	-7.10	-87.9
2.2	0.353	-9.05	-103.2
3.0	0.221	-13.10	-124.2
5.0	0.0875	-21.16	-148.0
8.0	0.0357	-28.96	-161.3

[Figures 3](#) and [4](#) show the measured frequency-response points together with the final model introduced later in [Section 3.3](#). The magnitude falls monotonically rather than showing a resonance peak. This is expected: a standard second-order low-pass system has a resonance peak only when $\zeta < 1/\sqrt{2} \approx 0.707$, whereas the step response already indicated a damping ratio near 0.75. Thus the lack of a resonance peak is consistent with, rather than contradictory to, the small time-domain overshoot.

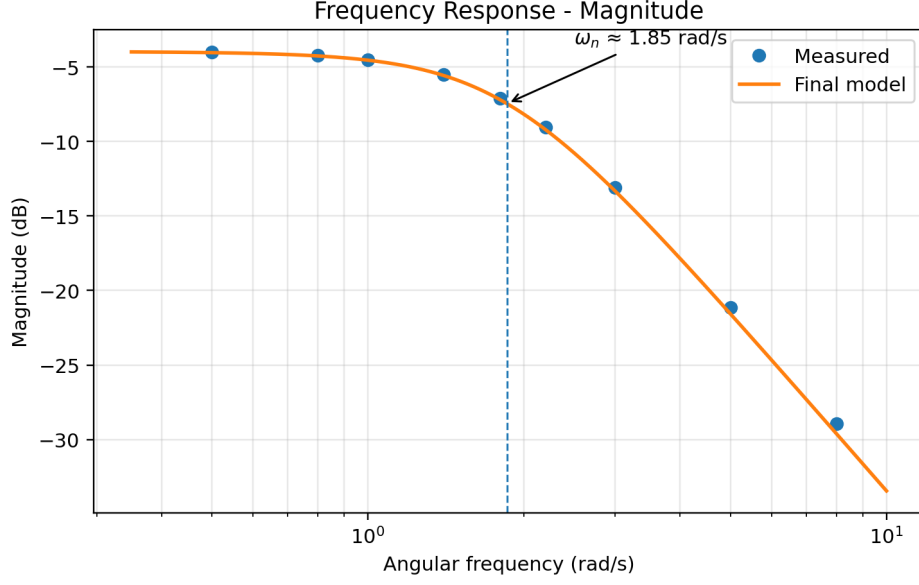


Figure 3: Measured frequency-response magnitude and final-model prediction.

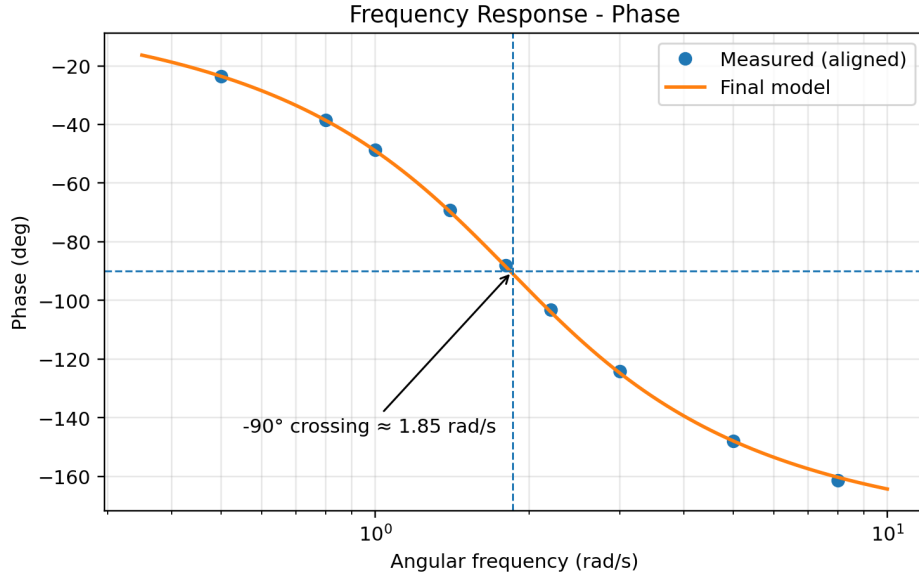


Figure 4: Aligned measured phase and final-model prediction. The -90 degree crossing occurs close to the natural frequency.

For a standard second-order low-pass model, the phase is exactly -90 degrees at $\omega = \omega_n$. Linear interpolation between the measured points at 1.8 and 2.200 rad s⁻¹ gives

$$\omega_{n,\text{freq}} \approx 1.854 \text{ rad s}^{-1}. \quad (20)$$

The lowest measured frequency gives the low-frequency gain estimate

$$K_{\text{freq}} \approx |G(j0.5)| = 0.631. \quad (21)$$

At $\omega = \omega_n$, a second-order model satisfies

$$|G(j\omega_n)| = \frac{K}{2\zeta}. \quad (22)$$

Interpolating the measured magnitude at the -90 degree crossing gives $|G(j\omega_n)| \approx 0.429$, and hence

$$\zeta_{\text{freq}} = \frac{K}{2|G(j\omega_n)|} = \frac{0.631}{2(0.429)} = 0.735. \quad (23)$$

The independently identified frequency-domain model is therefore

$$G_{\text{freq}}(s) = \frac{2.170}{s^2 + 2.726s + 3.437}, \quad (24)$$

with poles

$$s = -1.363 \pm j1.257. \quad (25)$$

These results closely confirm the gain, damping ratio, natural frequency and pole locations obtained from the step response.

3.3 Proposed System Model

Table 6 compares the independent estimates. All key parameters agree closely, with the natural-frequency estimates differing by about 2.3% and the damping-ratio estimates by about 1.4% relative to their average. A final model was therefore formed by averaging K , ζ and ω_n from the two experiments.

Table 6: Comparison of identified parameters.

Estimate	K	ζ	ω_n (rad/s)
Step test	0.638	0.745	1.813
Frequency test	0.631	0.735	1.854
Final average	0.635	0.740	1.833

The proposed model is

$$G(s) = \frac{X(s)}{U(s)} = \frac{2.133}{s^2 + 2.715s + 3.361} \quad (26)$$

with poles

$$s = -1.357 \pm j1.233. \quad (27)$$

Using Equation (5), the final spring and damping estimates (scaled to $m = 1$) are

$$b \approx 2.715, \quad k \approx 3.361. \quad (28)$$

The numerator $c \approx 2.133$ represents the proportionality between the simulator input and the applied-force term in the normalized model.

For convenient reference, the final model parameters are collected in Table 7.

Table 7: Final proposed second-order model parameters.

Quantity	Symbol	Final estimate
DC gain	K	0.635
Damping ratio	ζ	0.740
Natural frequency	ω_n	1.833 rad/s
Damping coefficient	b	2.715
Spring coefficient	k	3.361
Input-force proportionality	c	2.133
Poles	$s_{1,2}$	$-1.357 \pm j1.233$

3.4 Model Validation

The final model was tested against both experiments. Figure 5 compares the measured zero-adjusted step response to the response predicted by Equation (26) for the same 4-unit input. The model reproduces the rise, small overshoot and settling behaviour closely. Over the first 12 seconds after the step, the root-mean-square displacement error is approximately 0.028, which is about 1.1% of the measured steady-state output change.

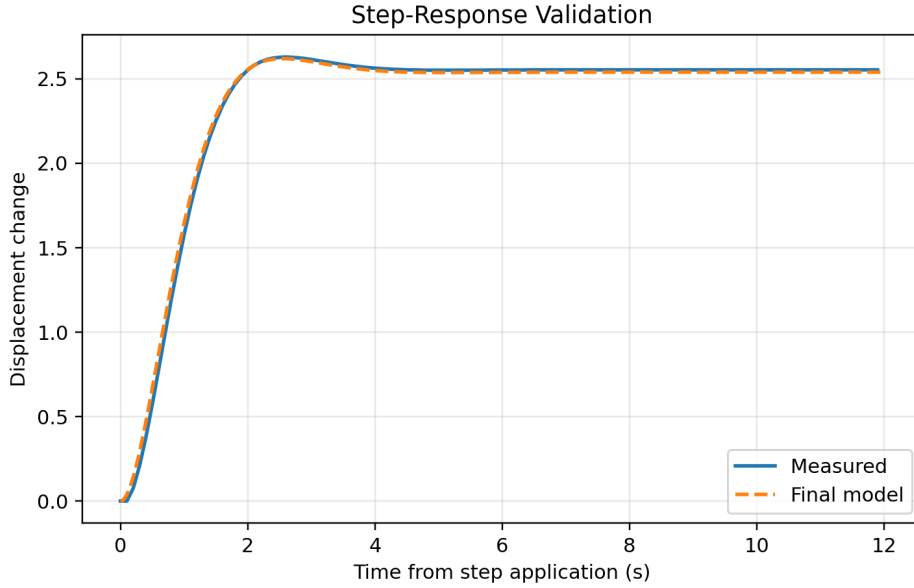


Figure 5: Validation of the proposed model against the measured step response.

The same model also tracks the measured Bode data in Figures 3 and 4. Across all nine tested frequencies, the RMS magnitude error is approximately 0.270 dB and the RMS phase error after sample-alignment correction is approximately 0.510 degree. The largest discrepancies are still below about 0.650 dB in magnitude and 1.000 degree in phase. The strong agreement in both domains indicates that a second-order model is sufficient to describe the dominant suspension dynamics over the measured range.

The validation statistics are summarised in Table 8.

Table 8: Summary of model-validation errors.

Validation quantity	Error measure	Result
Step response	RMS displacement error	0.028
Step response	RMS error relative to steady-state change	1.1%
Bode magnitude	RMS error	0.27 dB
Bode magnitude	Maximum error	< 0.65 dB
Bode phase	RMS error	0.51 deg
Bode phase	Maximum error	< 1 deg

4 Conclusion

A second-order model of the suspension simulator was successfully identified from independent step- and frequency-response tests. The independent estimates are compared in Table 6, and the final parameter set is summarised in Table 7. The step response had a steady-state gain of approximately 0.638, a small overshoot of 2.98%, and a peak time of 2.600 s, giving $\zeta \approx 0.745$, $\omega_n \approx 1.813 \text{ rad s}^{-1}$ and poles at approximately $-1.351 \pm j1.208$. The frequency test independently produced $K \approx 0.631$, $\zeta \approx 0.735$ and $\omega_n \approx 1.854 \text{ rad s}^{-1}$, confirming the time-domain result.

Averaging the two parameter sets gave the proposed transfer function

$$G(s) = \frac{2.133}{s^2 + 2.715s + 3.361},$$

corresponding to $b \approx 2.715$ and $k \approx 3.361$ for the normalized $m = 1$ system. The model closely reproduced both the measured step response and all nine frequency-response points. The absence of a resonant peak in the Bode magnitude is also consistent with the estimated damping ratio being slightly greater than $1/\sqrt{2}$. Overall, the identified transfer function provides a simple and well-validated representation of the dominant dynamics of the experimental system.

A Appendix and Notes

A.1 Measured Frequency-Response Values

The complete processed frequency-response data are supplied as `frequency_results.csv` in the accompanying source folder. [Table 5](#) contains the values used directly in the report.

A.2 MATLAB Validation Code

The following MATLAB code reproduces the final model and provides a direct step- and frequency-domain validation workflow. The measured Bode points in this listing are the values obtained from the uploaded experimental CSV files.

Listing 1: MATLAB code for model validation.

```
1 % EEE3094S Lab 1 - final model validation
2 clear; clc; close all;
3
4 % Final identified model
5 num = 2.13334;
6 den = [1 2.71463 3.36149];
7 G = tf(num, den);
8
9 % Step validation: experimental test used a 4-unit step
10 T = readtable('data/F34BSuspensionTestData.CSV');
11 t = T('Time(s)');
12 u = T.Input;
13 y = T.Output_Displacement;
14
15 idx = find(diff(u) ~= 0, 1, 'first') + 1;
16 t0 = t(idx);
17 y0 = mean(y(max(1,idx-50):idx-1));
18 tr = t(idx:end) - t0;
19 y_meas = y(idx:end) - y0;
20
21 [y_model, tm] = step(4*G, tr);
22 figure;
23 plot(tr, y_meas, 'LineWidth', 1.4); hold on;
24 plot(tm, y_model, '--', 'LineWidth', 1.4);
25 grid on; xlabel('Time (s)'); ylabel('Displacement change');
26 legend('Measured','Model');
27 title('Step-response validation');
28
29 % Measured frequency-response points
30 w = [0.5 0.8 1.0 1.4 1.8 2.2 3 5 8];
31 mag_dB = [-3.996 -4.221 -4.506 -5.516 -7.105 ...
32           -9.052 -13.101 -21.155 -28.957];
33 phase_deg = [-23.521 -38.443 -48.735 -69.279 -87.936 ...
34             -103.230 -124.197 -147.966 -161.258];
35
36 [mag, phase] = bode(G, w);
37 mag_model_dB = 20*log10(squeeze(mag));
```

```

38 phase_model = squeeze(phase);
39
40 figure;
41 semilogx(w, mag_dB, 'o', w, mag_model_dB, '-');
42 grid on; xlabel('Frequency (rad/s)'); ylabel('Magnitude (dB)');
43 legend('Measured','Model');
44 title('Frequency-response magnitude validation');
45
46 figure;
47 semilogx(w, phase_deg, 'o', w, phase_model, '-');
48 grid on; xlabel('Frequency (rad/s)'); ylabel('Phase (deg)');
49 legend('Measured','Model');
50 title('Frequency-response phase validation');

```

A.3 Raw Data

The accompanying source archive includes the original step-test CSV and all nine original frequency-test CSV files so that every result in the report can be reproduced from the raw measurements.

References

- [1] EEE3094S Control Systems Engineering, *Lab 1: System Identification*, University of Cape Town, 2026.
- [2] UCT Control Laboratory, *Control Laboratory Suspension Simulator User Manual*, Release 1.0, 10 Aug. 2023.